A

BRAC UNIVERSITY
Department of Computer Science Engineering
Mid Term Examination, Fall 2023
CSE 350: Digital Electronics and Pulse Techniques

Total Marks: **60 (20 x 3)**
Time: **60 Minutes**

- Total no. of questions are **four (4)**. Answer any **three (3)**.
- Figure in bracket [ ] next to each question indicates marks for that question.
- Use $\boldsymbol{V_\gamma(diode) = 0.6\ V}$, $\boldsymbol{V_\gamma(transistor) = 0.5\ V}$, $\boldsymbol{V_{BE}(forward\ active) = 0.7\ V}$ $\boldsymbol{V_D(conducting\ voltage\ of\ Diode) = 0.7\ V}$, for all the questions.

Name:	ID:	Section:

Question 1 [20]

An RTL circuit is given below and β_F for the transistor used here is equal to 40. The input voltage, V_{in} of the circuit can be in either of these two states: HIGH means **5V** and LOW means **-10V**. Here, R_2 = **(50 + 0.1*XY)** kΩ where, XY=last two digits of your ID.

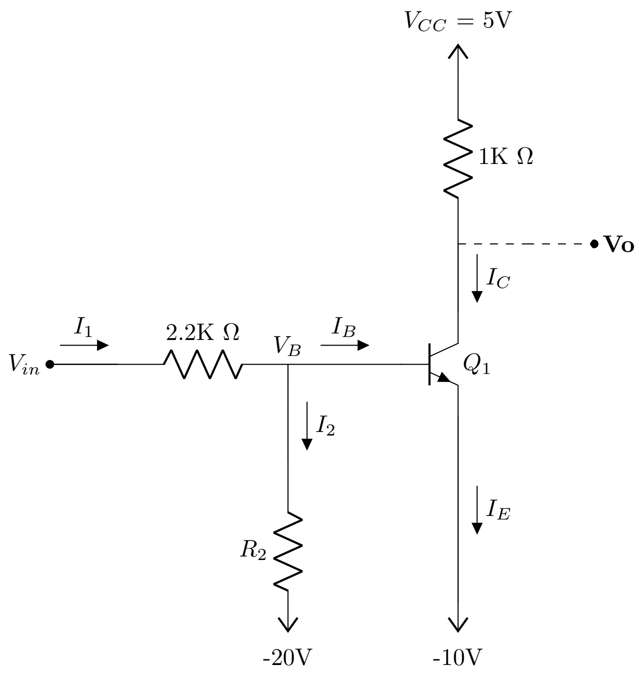

Figure 1

CO1	(a)	Find the voltages V_B and V_0 for each input combination.	[8]
CO1	(b)	If an identical RTL load circuit is connected to the output of the above circuit, **find** the emitter current i_E of the driver. [here, V_{in} is in HIGH state]	[12]

Question 2 **[20]**

For the circuit shown below, there are total 4 input terminals marked as V1, V2 ,V3 and V4. The input voltages can be either 5V (HIGH state) and 0V (LOW state). The junction voltage for each of the diode is 0.7 V. Here **XYZ=last 3 digits of your student ID.**

CO1	(a)	**Calculate** the power dissipated by all three resistors when V1, V3 = 5V and V2, V4 = 0V. [Hint: to calculate power dissipated by each resistor, calculate the current through that resistor and use formula $P = I^2R$]	[6]
CO1	(b)	**Find** the input voltage combination of V1, V2, V3 and V4 in V, for which D5 diode turns on and D6 diode turns off. [you can write the answer in one line, for example : V1 = 5V, V2= 5V, V3= 5V and V4=5V.]	[8]
CO1	(c)	**Determine** the boolean expression of V_{final} in terms of input V1, V2, V3 and V4. [Use (.) for AND and (+) for OR].	[6]

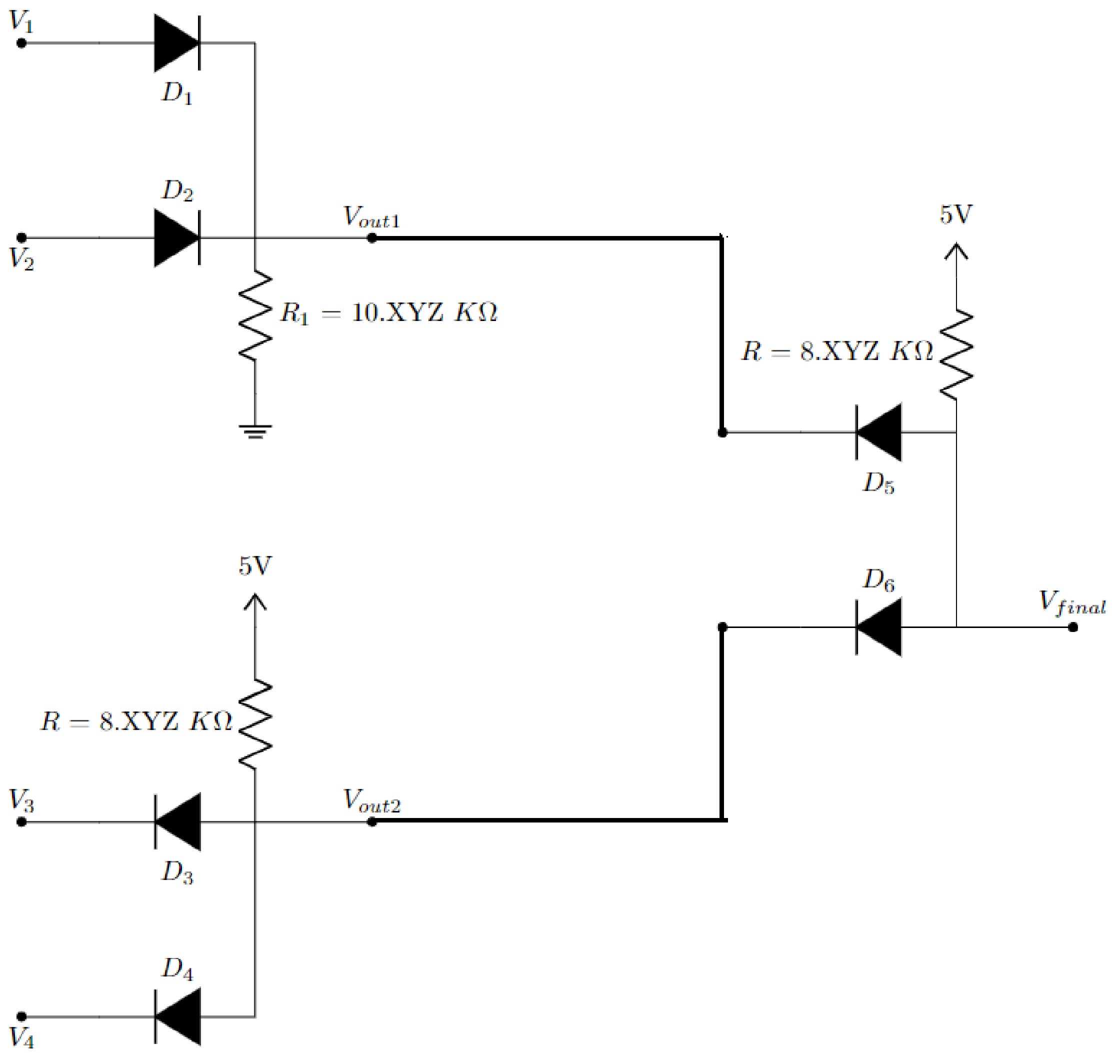

Figure 2

Question 3 [20]

For the DTL circuit in Figure 3 assume $\beta_F = 60 + X$, where **X=last digit of your ID**, $V_{BE(saturation)} = 0.8V$, and $V_{CE(saturation)} = 0.2V$. The input voltages 'A' and 'B' can be either 0.2V or 12V.

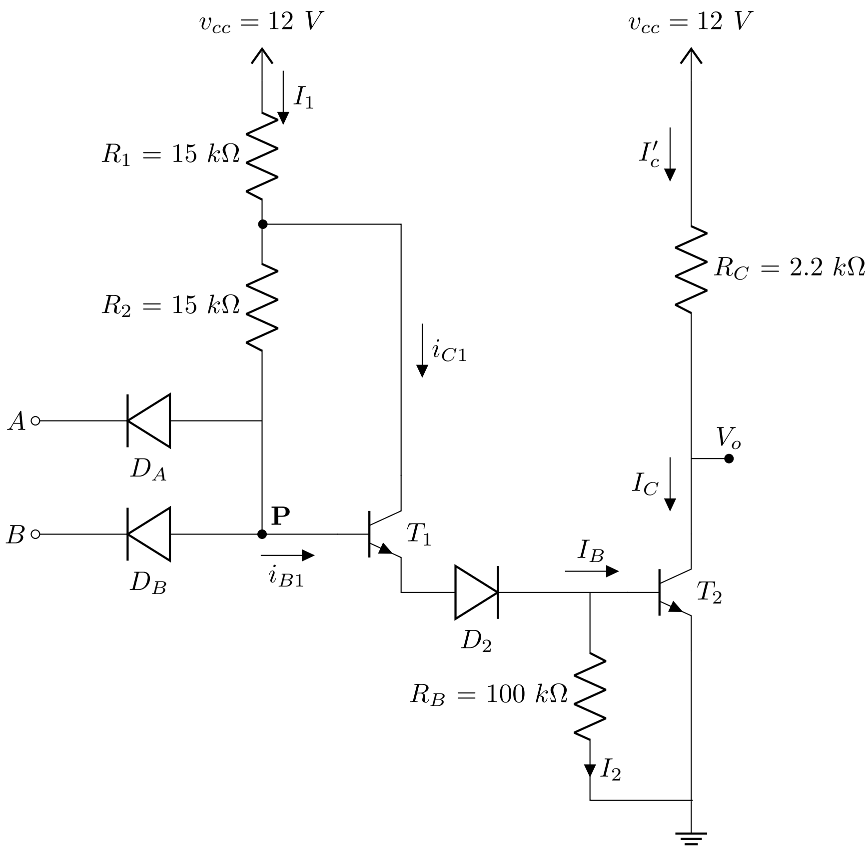

Figure 3

CO2	(a)	**Determine** the power dissipation for the DTL NAND gate when both the inputs 'A' and 'B' are LOW.	[4]
CO2	(b)	**Identify** what should be the minimum current gain, β_{min} to prevent the circuit from malfunctioning. [Hint: Consider both inputs as high]	[7]
CO2	(c)	**Estimate** the maximum fanout this logic device can handle without any issues.	[9]

Question 4 [20]

For the given TTL circuit in Figure 4, **common emitter current gain,** $\boldsymbol{\beta_F = 30}$, $\boldsymbol{V_{BE(sat)} = 0.8V}$, $\boldsymbol{V_{CE(sat)} = 0.1V}$, and **reverse common emitter current gain of the transistors** $\boldsymbol{\beta_R = (0.1 + 0.1 * X)}$**,** where X=last digit of your ID.

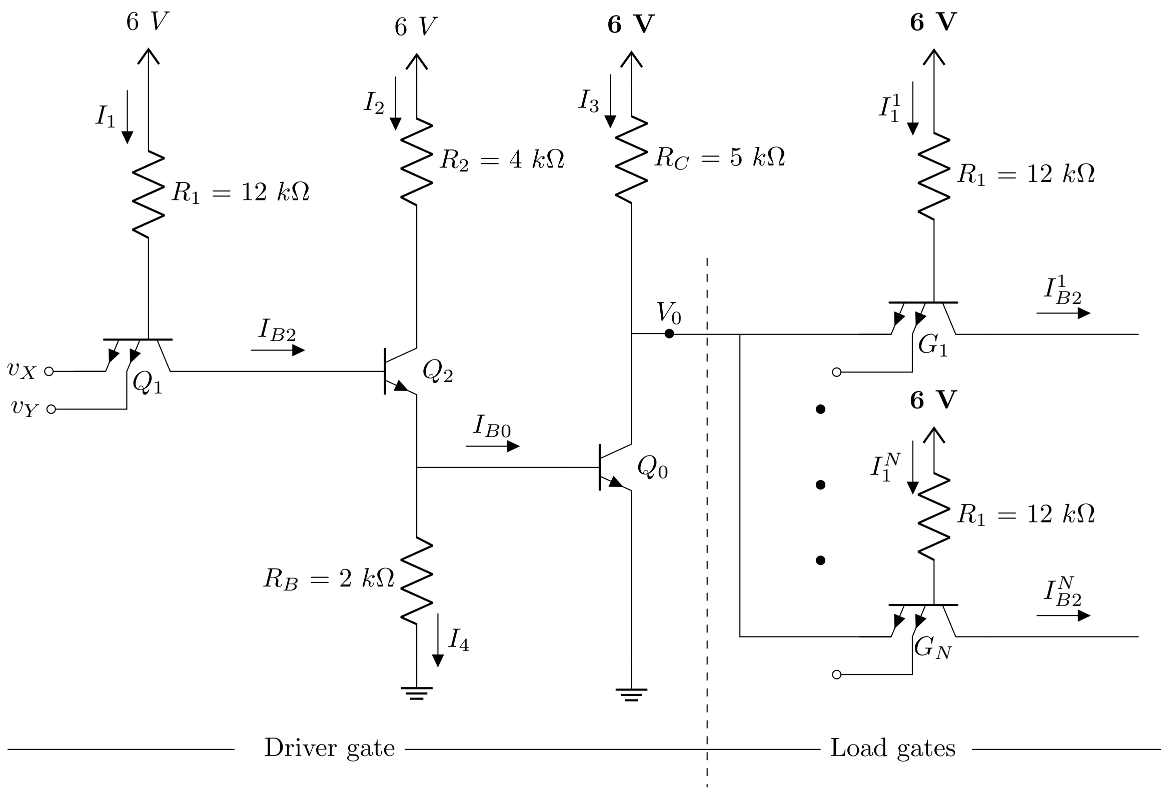

Figure 4

CO2	(a)	(i) **Determine** the maximum possible fanout when driver output is low (0.1V). (ii) If both the inputs of the load circuit are low, how the fanout will change in this case?	[10+4]
CO2	(b)	**Find** the total **power dissipation** in the driver circuit under the load configuration drawn in the figure.	[6]